Factorization memory in a conic ideal: the A_3, B_3, and H_3 configurations

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Abstract

Restriction to a conic forgets how its marked points were paired into secants. We study the information retained after differences of matching products are divided by the conic equation. We prove that this quotient is intrinsic up to translation and equivariant under the endpoint stabilizer. For the rank-three Coxeter configurations A_3, B_3, and H_3, its difference space has ranks 3, 6, and 10. Uniformly, the image contains the top harmonic summand and, in even degree, the deepest radial line. In the H_3 case it omits exactly the middle Fischer layer $Q\mathcal{H}_2$. In the balanced cases, second moments recover two complementary sheets, a signed cubic orients them, and six incidence profiles recover the matching-table row. A modular depth quotient and arithmetic gluing describe how this factorization memory changes across characteristics.

Keywords. conic ideal; secant matching; Coxeter configuration; factorization; cubic invariant; modular representation.

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1 Introduction

The companion rigidity paper isolates the Clebsch hexagon through its uncovered syndrome locus and reconstructs its projective geometry from decoding data. Here we begin with marked secant geometry and ask a complementary reconstruction question: what does common restriction to a conic forget, and what does the conic-ideal quotient remember? We answer this question across the rank-three A_3, B_3, and H_3 configurations.

Theorem 1.1 (Factorization recovery)**.** *Let the A_3, B_3, and H_3 matching configurations be the finite matching orbits defined in Section 4. Then:*

(i) *canonically scaled matching products have equal restriction to the conic, and their conic-ideal quotients have difference ranks* 3*,* 6*, and* 10*;*

(ii) *in the B_3 and H_3 cases, the two factorization sheets are, up to interchange, the unique complementary halves with equal first and second tensor moments;*

(iii) *the signed moments vanish through degree two, while a nonzero degree-three tensor transforms by the character that exchanges the sheets;*

(iv) *in the H_3 case, six explicit incidence profiles separate the six A_4–A_5 double-coset labels, and the singleton profiles recover the corresponding decorated matching-table row.*

The theorem follows one mechanism from a four-endpoint switch to the recovery of finite matching data. Division by the conic equation first turns different factorizations of one restricted form into an affine configuration. Quadratic products then recover an unordered bipartition; the first signed moment that survives is cubic. Finally, a double-coset incidence map compresses the configuration to six profiles without identifying any two labels. The rigidity theorem motivates this question but is not a hypothesis: all marked-conic objects and quotient constructions are defined independently here.

This line of research arose from the following normal-play placement game. Two players alternately adjoin a point of a finite projective plane while keeping the selected set an arc; the first player with no legal point loses. After frame reduction in an odd projective plane, a residual three-point position and the two burned directions determine a five-arc and hence a conic; adjoining an on-conic response exposes the exceptional six-arc geometry studied here. The resulting connection now runs in both directions. The small-arc equality theorem isolates the projective frame over $\mathbb{F}_5$ and the Clebsch hexagon over $\mathbb{F}_{11}$ as the only configurations of sizes four through seven whose uncovered locus is a full conic. The latter yields an icosahedral continuation complex, a certified game position, and a symmetry-compressed finite model for constructing second-player responses. Under the standard arc–MDS–AME dictionaries, the same H_3 six-arc gives an $[6,3,4]_{11}$ MDS code and a minimum-support stabilizer absolutely maximally entangled state on six 11-level systems, so its projective rigidity controls a concrete quantum-code input. More generally, prescribed-hole arc theory places its equality and reconstruction properties in a forward/inverse framework, while the syndrome–uncovered-point mechanism extends to projective Reed–Solomon deep holes beyond redundancy four. These are applications and continuations of the geometry, not hypotheses of the results proved here.

2 Conic matching products

Write the standard conic as

$$\mathcal{C} = V(Q), \qquad Q = XZ - Y^2,$$

and parametrize it by $\nu(s:t) = (s^2 : st : t^2)$. For points $a = (a_0 : a_1)$ and $b = (b_0 : b_1)$ of $\mathbb{P}^1$, put $[a,b] = a_0b_1 - a_1b_0$. We scale the secant L_{ab} through $\nu(a)$ and $\nu(b)$ by the condition

$$L_{ab}(\nu(s:t)) = [a,(s:t)]\,[b,(s:t)].$$

Let Ω be a set of $2m$ distinct marked points of $\mathbb{P}^1$, with fixed vector representatives. For a perfect matching M of Ω, define its matching product

$$P_M = \prod_{\{a,b\}\in M} L_{ab}.$$

The vector representatives are part of the marked-conic data. Their role is to make the products comparable before passing to projective equivalence.

Proposition 2.1 (The matching-secant quotient)**.** *For every $m \geq 2$ and every two perfect matchings M, N of Ω,*

$$P_M|_{\mathcal{C}} = P_N|_{\mathcal{C}}, \qquad P_M - P_N \in (Q).$$

Consequently, after choosing a reference matching M_0,

$$\Phi_{M_0}(M) = \frac{P_M - P_{M_0}}{Q} \in H^0(\mathbb{P}^2, \mathcal{O}(m-2)) \tag{2.1}$$

is an affine configuration. Changing the reference matching translates this configuration. If $g \in \mathrm{GL}_2$ preserves Ω, $\rho(g)$ is its action on binary quadratics, and representatives are transported by g, then

$$\Phi_{gM_0}(gM) \circ \rho(g) = \det(g)^{2m-2}\Phi_{M_0}(M). \tag{2.2}$$

Thus the difference space, its rank, and its affine moment conditions do not depend on M_0 and are equivariant under the endpoint stabilizer.

Proof. For $x \in \mathbb{P}^1$,

$$P_M(\nu(x)) = \prod_{a\in\Omega}[a,x],$$

which does not depend on the pairing. The kernel of the Veronese pullback

$$k[X,Y,Z] \longrightarrow k[s,t], \qquad (X,Y,Z) \longmapsto (s^2, st, t^2)$$

is the principal ideal (Q), so Q divides every difference. Changing the reference follows by subtraction. Finally, $L_{ga,gb} \circ \rho(g) = \det(g)^2 L_{ab}$ and $Q \circ \rho(g) = \det(g)^2 Q$, which gives (2.2). □

3 Switches and divisibility

For four distinct endpoints, the Plücker relation gives the local switch identity

$$L_{ab}L_{cd} - L_{ac}L_{bd} = [a,d][b,c]\,Q. \tag{3.1}$$

Multiplying (3.1) by the secant factors common to two matchings proves divisibility for a single switch. Any two perfect matchings are joined by a sequence of such switches: if M and N differ at a, write $\{a,b\} \in M$ and $\{a,c\} \in N$; switching the two M-edges through b and c creates $\{a,c\}$ and reduces the symmetric difference. This also gives an elementary proof of the divisibility clause in Proposition 2.1.

The switch calculation and the global pullback proof play different roles. The pullback proves the quotient for arbitrary matchings at once. The switch identity records how local changes of factorization move in the quotient space. A selected Coxeter orbit need not contain every perfect matching or every intermediate switch.

4 The rank-three configurations

We now select three finite matching orbits. This selection is discrete input: Proposition 2.1 applies to every matching, but it does not distinguish the Coxeter orbits.

For $q \in \{5, 7, 11\}$, index

$$\mathbb{P}^1(\mathbb{F}_q) = \{0, 1, \ldots, q-1, \infty\},$$

where a denotes $(1 : a)$ and ∞ denotes $(0 : 1)$. Let $G_q = \mathrm{PGL}_2(q)$ act in the usual way. The three base matchings are

T	q	M_T
A_3	5	$\{0,\infty\}, \{1,4\}, \{2,3\}$,
B_3	7	$\{0,2\}, \{1,4\}, \{3,\infty\}, \{5,6\}$,
H_3	11	$\{0,1\}, \{2,5\}, \{3,7\}, \{4,9\}, \{6,8\}, \{10,\infty\}$.

(4.1)

Define the *rank-three matching configuration*

$$\Omega_T = G_q \cdot M_T.$$

The stabilizers of the three displayed matchings are respectively S_4, S_4, A_5, and hence

$$|\Omega_{A_3}| = 5, \qquad |\Omega_{B_3}| = 14, \qquad |\Omega_{H_3}| = 22. \tag{4.2}$$

These marker spaces and their stabilizers are classical. In particular, Edge and Dye identify the exceptional finite conic configurations and substantial parts of their incidence geometry [1, 2]. We use the Coxeter names for the corresponding tetrahedral, octahedral, and icosahedral data; neither the orbit counts in (4.2) nor the ambiguity of an unmarked conic is asserted here as new.

Fix M_T as reference and apply (2.1) to every member of Ω_T. Write $R_d = \mathbb{F}_q[X, Y, Z]_d$ and

$$\Delta_Q = 4\partial_X\partial_Z - \partial_Y^2, \qquad \mathcal{H}_d = \ker(\Delta_Q : R_d \longrightarrow R_{d-2}). \tag{4.3}$$

The operator Δ_Q is the Laplacian attached to the dual quadratic form. In the degrees used below, $\mathcal{H}_d$ is the top harmonic summand, while powers of Q give radial summands.

Theorem 4.1 (The $3, 6, 10$ quotient ranks)**.** *For the matching configurations in (4.1), let*

$$W_T = \operatorname{span}_{\mathbb{F}_q}\{\Phi_{M_T}(M) : M \in \Omega_T\}.$$

Then

$$\begin{array}{cccc} T & \deg\Phi & W_T & \dim W_T \\ \hline A_3 & 1 & R_1 = \mathcal{H}_1 & 3, \\ B_3 & 2 & R_2 = \mathcal{H}_2 \oplus \mathbb{F}_7 Q & 6, \\ H_3 & 4 & \mathcal{H}_4 \oplus \mathbb{F}_{11} Q^2 & 10. \end{array} \tag{4.4}$$

If $h_T = 4, 6, 10$ is the Coxeter number and $d = h_T/2 - 1$, the three rows have the uniform form

$$W_T = \begin{cases} \mathcal{H}_d, & d \text{ odd}, \\ \mathcal{H}_d \oplus \mathbb{F}_q Q^{d/2}, & d \text{ even}, \end{cases} \qquad \dim W_T = h_T - 1 + \mathbf{1}_{\{d \text{ even}\}}.$$

In particular, the first two configurations fill the complete quotient form space. The H_3 configuration spans a canonical ten-dimensional subspace of the fifteen-dimensional space R_4.

Proof mode. The quotient construction and the harmonic-space dimensions are proved symbolically. The assertions that the three displayed finite orbits span the stated subspaces are certificate-checked exact calculations, with a separate implementation that reconstructs the groups, matchings, quotient forms, Laplacian, and row reductions from the data in (4.1).

Proof. A matching in (4.1) has $m = 3, 4, 6$ edges, respectively, so Proposition 2.1 places its quotient in R_{m-2}, of degrees $1, 2, 4$. Representing the quotients in the standard monomial bases gives matrices with respectively $5, 14, 22$ rows. Exact row reduction over $\mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$ gives ranks $3, 6, 10$.

It remains to identify the ten-space in the last row rather than record only its dimension. Direct differentiation gives

$$\dim \mathcal{H}_1 = 3, \qquad \dim \mathcal{H}_2 = 5, \qquad \dim \mathcal{H}_4 = 9.$$

The radial vectors $Q \in R_2$ and $Q^2 \in R_4$ are not harmonic in characteristics 7 and 11, respectively. The exact quotient matrices lie in

$$\mathcal{H}_1, \qquad \mathcal{H}_2 + \mathbb{F}_7 Q, \qquad \mathcal{H}_4 + \mathbb{F}_{11} Q^2.$$

Their computed ranks equal the dimensions $3, 6, 10$ of these spaces, so all three containments are equalities. Since $\dim R_4 = 15$, the last equality is proper. $\square$

Corollary 4.2 (The exact H_3 loss)**.** *Over $\mathbb{F}_{11}$, the quartic form space has the Fischer decomposition*

$$R_4 = \mathcal{H}_4 \oplus Q\mathcal{H}_2 \oplus \mathbb{F}_{11} Q^2. \tag{4.5}$$

Consequently,

$$R_4 = W_{H_3} \oplus Q\mathcal{H}_2, \qquad R_4 / W_{H_3} \simeq Q\mathcal{H}_2. \tag{4.6}$$

Equivalently,

$$W_{H_3} = \{f \in R_4 : \Delta_Q f \in \mathbb{F}_{11} Q\}. \tag{4.7}$$

Thus the H_3 quotient configuration retains the top harmonic nine-space and the deepest radial line, and omits precisely the five-dimensional middle radial layer.

Proof. For a homogeneous form f of degree r, direct differentiation gives

$$\Delta_Q(Qf) = (4r + 6)f + Q\Delta_Q f. \tag{4.8}$$

Hence $\Delta_Q(Qh) = 3h$ for $h \in \mathcal{H}_2$, while $\Delta_Q(Q^2) = 9Q$ over $\mathbb{F}_{11}$. Applying Δ_Q^2 and then Δ_Q shows that the three summands in (4.5) meet trivially. Their dimensions are $9, 5, 1$, so they exhaust the fifteen-dimensional space R_4. Theorem 4.1 then gives (4.6). If $f = h_4 + Qh_2 + cQ^2$ is its decomposition, then

$$\Delta_Q f = 3h_2 + 9cQ.$$

The quadratic decomposition $R_2 = \mathcal{H}_2 \oplus \mathbb{F}_{11} Q$ therefore gives (4.7). $\square$

The rank calculation has two distinct boundaries. The common restriction and quotient are general consequences of the conic ideal, and (4.3) describes the intrinsic target subspaces. The fact that the three particular orbits in (4.1) attain those subspaces is finite. The primary certificate enumerates each full orbit and performs exact prime-field row reduction; the independent replay starts only from (4.1) and reconstructs the calculation without importing the primary implementation. Thus Theorem 4.1 does not infer a general rank formula from the three examples.

Equation (4.7) isolates a smaller conceptual target than the full rank calculation: the H_3 containment is the vanishing of the harmonic quadratic part of $\Delta_Q \Phi(M)$. Once that covariant identity is known, only spanning the resulting ten-space remains finite.

Corollary 4.2 identifies the linear information lost by the H_3 orbit, but linear span does not recover the two factorization sheets inside the finite configuration. Their recovery begins one categorical level higher, with products of affine evaluation functions.

5 Balanced sheets and cubic orientation

For $T = B_3, H_3$, put $q = 7, 11$, respectively. The subgroup

$$G^+ = \mathrm{PSL}_2(q) \subset G = \mathrm{PGL}_2(q)$$

splits Ω_T into two orbits $\Omega_+ \sqcup \Omega_-$, each of size q; an element of the outer coset exchanges them. These are the two one-factorization sheets. The A_3 orbit does not split: its S_4 stabilizer cannot lie in $\mathrm{PSL}_2(5) \simeq A_5$, already by order.

Write $x_M = \Phi_{M_T}(M) \in W_T$. Let $L \subseteq \mathbb{F}_q^{\Omega_T}$ be the evaluation space of affine-linear functions on the finite configuration $\{x_M\}$. Since the points affinely span W_T,

$$\dim L = 1 + \dim W_T = q.$$

For subspaces of functions, write

$$L^{\circ 2} = \mathrm{span}\{fg : f, g \in L\},$$

where multiplication is pointwise.

Proposition 5.1 (Radical–Hadamard recovery)**.** *Let k be a field and $\Omega = \Omega_+ \sqcup \Omega_-$, with both sheets of size $q = 0$ in k. Suppose an affine evaluation space $L \subseteq k^{\Omega}$ has the following properties:*

(i) *each sheet has zero first moment, and the two sheets have equal second moments;*

(ii) *$\dim L = q$, and restriction from L onto the zero-sum hyperplane of either sheet has rank $q - 1$;*

(iii) *the second-moment form on linear functionals has rank $q - 2$ and a one-dimensional radical;*

(iv) *a nonzero radical functional is constant on each sheet, with different constants on the two sheets.*

Then

$$L^{\circ 2} = \left\{ (u_+, u_-) : \sum_{\Omega_+} u_+ = \sum_{\Omega_-} u_- \right\}, \tag{5.1}$$

and

$$(L^{\circ 2})^{\perp} = k(\mathbf{1}_{\Omega_+} - \mathbf{1}_{\Omega_-}). \tag{5.2}$$

Consequently every field-valued strength-two trade is a scalar multiple of the sheet sign.

Proof. Let e_+ and e_- be the two sheet indicators. The radical functional evaluates as $a_+ e_+ + a_- e_-$ with $a_+ \neq a_-$. Together with $1 = e_+ + e_- \in L$, it therefore yields $e_+, e_- \in L$.

Let $H_\pm$ be the zero-sum hyperplane on $\Omega_\pm$. The first-moment hypothesis and $q = 0$ put each restriction of L inside $H_\pm$; the rank hypothesis makes both restrictions surjective. Multiplication by e_+ and e_- now gives

$$(H_+ \oplus 0) + (0 \oplus H_-) \subseteq L^{\circ 2}. \tag{5.3}$$

This subspace has dimension $2q - 2$.

For $f, g \in L$, the difference between the two sheet sums of fg expands into constant, first-moment, and second-moment terms. The hypotheses make all three differences zero, so $L^{\circ 2}$ lies in the hyperplane on the right of (5.1). Finally choose restrictions $a, b \in H_+$ with nonzero standard pairing and lift them to $f, g \in L$. Equality of the second moments gives the same nonzero pairing on Ω_-. Thus fg has equal nonzero sheet sums and does not lie in (5.3). We have found $2q - 1$ independent directions in the $2q - 1$-dimensional hyperplane, proving (5.1). Its annihilator under the standard coordinate pairing is the line in (5.2). □

The hypotheses of Proposition 5.1 are small finite linear-algebra statements for the two quotient configurations:

T	q	sheet restriction ranks	second-moment rank	$\dim L^{\circ 2}$
B_3	7	$6, 6$	5	13,
H_3	11	$10, 10$	9	21.

(5.4)

In each row the radical is one-dimensional and takes two distinct sheet values. The exact certificate checks these hypotheses from the matching quotients; an independent implementation reconstructs the row reductions and the resulting trade line. No enumeration of equal-size subsets is used in the proof.

Theorem 5.2 (Balanced sheets and cubic-first orientation)**.** *For $T = B_3, H_3$, the unordered pair $\{\Omega_+, \Omega_-\}$ is intrinsic to the affine quotient configuration:*

(i) *up to interchange, Ω_+ and Ω_- are the only complementary halves with equal first and second tensor moments;*

(ii) *if ϵ is $+1$ on Ω_+ and -1 on Ω_-, then*

$$\mu_j = \sum_{M \in \Omega_T} \epsilon(M) x_M^{\odot j} \in \mathrm{Sym}^j(W_T)$$

satisfies

$$\mu_1 = 0, \qquad \mu_2 = 0, \qquad \mu_3 \neq 0; \tag{5.5}$$

(iii) *μ_3 is independent of the reference matching, G^+ fixes it, and the outer coset negates it. Within G,*

$$\mathrm{Stab}_G(\mu_3) = G^+, \qquad \mathrm{Stab}_G(\mathbb{F}_q \mu_3) = G. \tag{5.6}$$

Thus degree three is the first signed tensor moment, or linear functional of such a moment, that orients the recovered sheet pair. This minimality statement concerns signed tensor moments, not every possible nonlinear statistic of the configuration.

Proof mode. Proposition 5.1 is a symbolic proof. The ranks, radical separation, lower-moment cancellation, and nonzero cubic for the two displayed configurations are certificate-checked over $\mathbb{F}_7$ and $\mathbb{F}_{11}$ and independently replayed. The transformation law and reference independence below are formal consequences of those finite inputs.

Proof. Equality of first and second tensor moments for a complementary pair is equivalent, by polarization, to its sign weight annihilating $L^{\circ 2}$. Since the characteristics are 7 and 11, polarization through degree two is valid. Proposition 5.1 says that the annihilator is exactly the sheet-sign line. A sign weight taking only the values $\{\pm 1\}$ is therefore ϵ or $-\epsilon$, proving (i).

The same moment equalities give $\mu_1 = \mu_2 = 0$. Exact tensor summation in the rank-six and rank-ten quotient coordinates gives $\mu_3 \neq 0$. If the reference matching changes, every point translates by one vector c. Expanding the signed third moment after $x_M \mapsto x_M + c$ leaves μ_3 unchanged: the other terms contain μ_2, μ_1, or $\mu_0 = \sum_M \epsilon(M) = 0$.

Every element of G^+ preserves both sheets and hence fixes the signed sum. Every element of the outer coset exchanges the sheets and negates it. Since $\mu_3 \neq 0$, these two transformation laws give (5.6). Finally (5.5) proves the stated minimality inside the signed tensor-moment family. □

The cubic remembers an orientation that the linear quotient does not. Indeed, the signed linear relation

$$\sum_{M\in\Omega_+} x_M = \sum_{M\in\Omega_-} x_M$$

already shows that the sheet sign is killed by the linear quotient map. Quadratic products recover the unordered partition through (5.2), and the cubic supplies the sign exchanged when the two recovered levels are swapped.

Corollary 5.3 (Secant-product tensor syzygies)**.** *For $T = B_3, H_3$, the canonically scaled secant products satisfy*

$$\begin{aligned} \sum_M \epsilon(M)P_M &= 0, \\ \sum_M \epsilon(M)P_M^{\odot 2} &= 0, \\ \sum_M \epsilon(M)P_M^{\odot 3} &= Q^{\odot 3}\mu_3 \neq 0. \end{aligned} \tag{5.7}$$

Proof. Substitute $P_M = P_{M_T} + Qx_M$ and expand. Each sheet has $q = 0$ elements in $\mathbb{F}_q$, and the signed quotient moments vanish through degree two. All terms below cubic degree cancel, leaving $Q^{\odot 3}\mu_3$ in degree three. □

6 Six profiles and matching-row reconstruction

We now specialize to H_3. Put

$$G = \mathrm{PGL}_2(11), \qquad G^+ = \mathrm{PSL}_2(11), \qquad X = \Omega_{H_3} \simeq G/H,$$

where $H \simeq A_5$ is the stabilizer of the base matching in (4.1). Thus $X = X_+ \sqcup X_-$, with $|X_+| = |X_-| = 11$. The ordered golden pair of A_5-parents selects a common subgroup $K \simeq A_4$ and an outer involution J which normalizes K and exchanges X_+ with X_-.

The nonzero scalar-K orbits in the ternary representation projectivize to a K-orbit refinement of the projective plane. The involution J exchanges four pairs of cells, which we denote

$$(R_1, R_{10}), \quad (R_3, R_{13}), \quad (R_6, R_{14}), \quad (R_9, R_{11}). \tag{6.1}$$

Only these four oriented pairs enter the argument. Reversing the golden pair reverses all four orientations.

This profile layer uses more decoration than the balanced-sheet theorem. The quotient configuration intrinsically recovers the unordered sheets, but it does not select an ordered golden pair or the resulting scalar-K refinement. The reconstruction below is relative to that selected refinement.

For $M \in X$, let S_M be the union of its six secants in $\mathbb{P}^2(\mathbb{F}_{11})$, and put

$$Z_M(R) = |S_M \cap R|.$$

Equivalently, $Z_M(R)$ counts the points of R at which P_M vanishes. Define the *depth profile*

$$\begin{aligned} D(M) = (Z_M(R_1) - Z_M(R_{10}),\ & Z_M(R_3) - Z_M(R_{13}), \\ & Z_M(R_6) - Z_M(R_{14}),\ Z_M(R_9) - Z_M(R_{11})) \in \mathbb{F}_{11}^4. \end{aligned} \tag{6.2}$$

This definition uses only zero sets of secant products, so it is unchanged if their equations are rescaled.

Theorem 6.1 (Six-profile reconstruction)**.** *The K-orbits on each sheet $X_\pm$ have sizes* $1, 4, 6$. *With the orientation in (6.1), the corresponding profiles and fibre sizes are*

$$\begin{array}{ccc} \textit{sheet} & |D^{-1}(v)| & v \\ \hline + & 1 & v_1 = (-6, 0, 12, -12), \\ + & 4 & v_2 = (-3, 3, 0, 3), \\ + & 6 & v_3 = (3, -2, -2, 0), \\ - & 1 & -v_1, \\ - & 4 & -v_2, \\ - & 6 & -v_3. \end{array} \tag{6.3}$$

The six vectors are pairwise distinct. Hence D recovers the double-coset row in

$$K\backslash G/H$$

of every matching. Its linear span nevertheless has dimension only two: over $\mathbb{F}_{11}$ it is the plane

$$2a + 2b + c = 0, \qquad 9a + 8b + d = 0. \tag{6.4}$$

The two singleton profiles recover their individual matchings. Under the matching-decorated parent correspondence, they therefore recover the corresponding individual H_3 parent. A profile in a fibre of size four or six recovers only that K-orbit, not an individual matching.

Proof mode. The subgroup-mark reduction and the implications from equivariance are conceptual. The six representative incidence rows in (6.5) below are certificate-checked exact counts and are reconstructed by an independent implementation. A companion certificate verifies that the matching decoration uniquely recovers its parent on the 22-point H_3 locus.

Proof. The permutation character of K on either eleven-point sheet has values

$$(11, 3, 2)$$

on elements of orders $1, 2, 3$. The marks of the transitive A_4-sets are

K-set	size	marks on orders $1, 2, 3$
K/K	1	$(1, 1, 1)$
K/V_4	3	$(3, 3, 0)$
K/C_3	4	$(4, 0, 1)$
K/C_2	6	$(6, 2, 0)$
$K/1$	12	$(12, 0, 0)$.

Comparing marks gives the unique nonnegative decomposition

$$X_+|_K \simeq K/K \ \sqcup \ K/C_3 \ \sqcup \ K/C_2.$$

The orbit sizes are therefore $1, 4, 6$; conjugation by J gives the same decomposition on X_-.

For $k \in K$, the transformation k preserves every cell in (6.1) and carries S_M to S_{kM}. Thus $D(kM) = D(M)$. The involution J exchanges the two cells in every oriented pair and carries S_M to S_{JM}, so

$$D(JM) = -D(M).$$

It remains to count one representative on each positive-sheet orbit. The exact incidence rows are

$$\begin{array}{cccc} \text{orbit size} & \mathrm{Stab}_K(M) & \text{zero counts in the four pairs of (6.1)} & D(M) \\ \hline 1 & A_4 & (0,6;\ 0,0;\ 12,0;\ 0,12) & v_1, \\ 4 & C_3 & (3,6;\ 3,0;\ 3,3;\ 6,3) & v_2, \\ 6 & C_2 & (3,0;\ 1,3;\ 2,4;\ 6,6) & v_3. \end{array} \tag{6.5}$$

The J-images supply the three negative rows. These six vectors are pairwise distinct, so their fibres are precisely the six K-orbits.

Row reduction over $\mathbb{F}_{11}$ gives rank two and the equations (6.4). This rank drop does not merge orbit labels: linear rank and set-theoretic separation are different assertions. Finally, a singleton fibre contains its matching without residual choice. The stabilizer of an obstruction matching is the same A_5 as the stabilizer of its parent. The equivariant decorated transform between the two G/H-orbits is therefore a bijection, so the matching also determines its parent. The stated limitation for the other four rows follows from their fibre sizes. □

Corollary 6.2 (Decorated sheet classifier)**.** *Relative to the orientation in (6.1), the profile $D(M)$ determines the sheet containing M. The antipodal singleton pair $\{v_1, -v_1\}$ recovers the unordered golden matching pair and hence the unordered golden parent pair; choosing the orientation selects one member.*

Proof. The positive and negative profile sets in (6.3) are disjoint, and the only singleton fibres are v_1 and $-v_1$. Apply the matching–parent bijection from Theorem 6.1. □

This pointwise classifier does not contradict the failure of a linear sheet sign in Section 5. The map D uses zero incidences with the selected scalar-K refinement; it is not a linear functional on the affine quotient configuration. Without that extra decoration, the quadratic and cubic moment statements remain the intrinsic recovery mechanism.

The profile compression also retains the cubic-first orientation from Section 5. The three positive vectors satisfy the integral weighted relation

$$v_1 + 4v_2 + 6v_3 = 0. \tag{6.6}$$

Corollary 6.3 (Orbit weights from profile rays)**.** *The three antipodal rational profile rays recover the orbit-size multiset $\{1, 4, 6\}$. They also recover the stabilizer orders $\{12, 3, 2\}$ inside $K \simeq A_4$.*

Proof. The three integral profile vectors span a plane over $\mathbb{Q}$, so their rational relation space is one-dimensional. Equation (6.6) is its primitive positive generator. Its coefficients recover the three orbit sizes without labels. Orbit–stabilizer then gives $12/1, 12/4, 12/6$. □

Lemma 6.4 (Three-ray cubic forcing)**.** *Let k have characteristic different from 2 and 3. Suppose u_1, u_2 form a basis of a two-dimensional k-space and $n_1, n_2, n_3 \in k^\times$ satisfy*

$$n_1u_1 + n_2u_2 + n_3u_3 = 0.$$

For the signed antipodal measure

$$\xi = \sum_{i=1}^{3} n_i(\delta_{u_i} - \delta_{-u_i}),$$

the moments of degrees one and two vanish, while the cubic moment is nonzero.

Proof. The relation kills the first moment, and antipodality kills every even moment. Write

$$u_3 = -\frac{n_1}{n_3}u_1 - \frac{n_2}{n_3}u_2.$$

In $\sum_i n_i u_i^{\odot 3}$, the coefficient of $u_1^{\odot 2} \odot u_2$ is

$$-\frac{3n_1^2 n_2}{n_3^2},$$

which is nonzero. The signed cubic is twice this tensor and is therefore nonzero as well. □

Corollary 6.5 (The mass-zero cubic flag)**.** *In Lemma 6.4, suppose also that* $n_1+n_2+n_3 = 0$*. Then its half-cubic factors as*

$$\frac{1}{2}M_3(\xi) = \sum_{i=1}^{3} n_i u_i^{\odot 3} = \frac{n_1 n_2}{(n_1+n_2)^2}(u_1-u_2)^{\odot 2} \odot ((2n_1+n_2)u_1 + (n_1+2n_2)u_2). \tag{6.7}$$

The two displayed lines are distinct. Hence the cubic intrinsically defines a doubled line and a residual simple line; its Hessian recovers the doubled line.

Proof. Since $n_3 = -(n_1+n_2)$, the weighted vector relation gives

$$u_3 = \frac{n_1 u_1 + n_2 u_2}{n_1 + n_2}.$$

Substitution and expansion give (6.7). The residual factor would equal the doubled factor only if $3(n_1+n_2) = 0$, which is impossible under the hypotheses. The Hessian of a binary cubic $L^2 R$, with L and R distinct, is a nonzero scalar multiple of L^2. □

If

$$\eta = \delta_{v_1} + 4\delta_{v_2} + 6\delta_{v_3} - \delta_{-v_1} - 4\delta_{-v_2} - 6\delta_{-v_3},$$

then its signed symmetric moments obey

$$M_j(\eta) = (1-(-1)^j)\sum_{i=1}^{3} n_i v_i^{\odot j}, \qquad (n_1,n_2,n_3) = (1,4,6). \tag{6.8}$$

Equation (6.6) kills the first moment, and antipodality kills every even moment. The first two profiles are independent over $\mathbb{F}_{11}$, since the determinant of their first two coordinates is $-18 \neq 0$. Lemma 6.4 therefore forces the cubic to be nonzero without a coordinate calculation. In the frozen normalization, its first coordinate is

$$2((-6)^3 + 4(-3)^3 + 6(3)^3) = 6 \neq 0 \quad \text{in } \mathbb{F}_{11}. \tag{6.9}$$

Thus the six-row quotient preserves the first nonzero signed moment even though its linear image is only a plane. It is a set-separating realization of the mixed K–H double-coset space, not a faithful linear quotient of its six-dimensional function space.

Remark 6.6 (The H_3 cubic flag). Here $1+4+6 = 0$ in $\mathbb{F}_{11}$, so Corollary 6.5 forces the double-line factorization. In the basis $e_1 = v_2, e_2 = v_3$ of the profile plane, the projective cubic is represented by

$$f(x,y) = x^3 + 7x^2y + 5xy^2 + 9y^3 = (x-y)^2(x-2y).$$

Its Hessian is

$$7x^2 + 8xy + 7y^2 = 7(x-y)^2.$$

Thus the cubic divisor intrinsically distinguishes a doubled line and a residual simple line, and the Hessian recovers the doubled line. This is an internal flag of the decorated profile plane. It does not supply a canonical map from the full relative-cubic space to that plane, nor does the six-profile quotient carry a descended G^+-action.

7 Modular depth and arithmetic gluing

Present the modular depth quotient, the primitive $1:4:6$ relation, and the arithmetic splitting/gluing theorem as one change-of-characteristic story. Put the relative-cubic Tate plane in an appendix unless it shortens this argument.

8 Verification architecture

Distinguish conceptual proofs, classical inputs, Lean statements, and finite certificates. State the coordinate-to-Coxeter identifications and external semantic bridges explicitly.

A The relative-cubic Tate plane

Record the two canonical constructions and the proved boundary between the relative-cubic plane and the modular depth plane.

Conclusion

State exactly which factorization data the conic ideal remembers and which pointed or global data it still forgets.

References

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